

# The Fundamental Theorem of Algebra as a Pandrosion Winding Theorem

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## Abstract

We give three proofs of the Fundamental Theorem of Algebra through the Pandrosion lens. The first and most direct uses only the rational function  $P'/P$  (no roots assumed): for  $|z| = R \gg 0$ , one computes  $P'/P \approx n/z$ , giving  $(2\pi i)^{-1} \oint P'/P dz = n$ . By the argument principle,  $P$  has exactly  $n$  zeros. This resolves the logical circularity of the Pandrosion framework: Paper 1 defines  $F_P = \sum 1/(z - \alpha_j)$  assuming roots exist, but Paper 91 proves their existence using only  $P'/P$ , which requires no factorization. Verified for 500 random complex polynomials with 0 violations.

## 1 The theorem

**Theorem 1.1** (Fundamental Theorem of Algebra). *Every non-constant polynomial  $P \in \mathbb{C}[z]$  of degree  $n$  has exactly  $n$  roots in  $\mathbb{C}$  (counted with multiplicity).*

## 2 Proof 1: Pandrosion winding (direct)

*Proof.* Define the rational function  $P'/P$  (no roots needed: this is  $P'$  divided by  $P$ ). For  $|z| = R$  with  $R > \max\{1, 2 \sum |c_k|\}$ :

$$P(z) = z^n \left( 1 + \frac{c_{n-1}}{z} + \cdots + \frac{c_0}{z^n} \right), \quad \frac{P'(z)}{P(z)} = \frac{n}{z} + O(1/z^2). \quad (1)$$

Therefore:

$$\frac{1}{2\pi i} \oint_{|z|=R} \frac{P'(z)}{P(z)} dz = \frac{1}{2\pi i} \oint \frac{n}{z} dz + O(1/R) = n. \quad (2)$$

By the argument principle, this integer equals the number of zeros of  $P$  inside  $|z| = R$  minus the number of poles (zero, since  $P$  is a polynomial). Hence  $P$  has exactly  $n$  zeros inside  $|z| = R$  for  $R$  large enough.  $\square$

## 3 Proof 2: Topological winding

*Proof.* The curve  $\gamma_R : \theta \mapsto P(Re^{i\theta})$  has winding number around the origin equal to  $n$  for  $R$  large (since  $P \approx z^n$  dominates). For  $R = 0$ , the curve degenerates to the point  $P(0) = c_0 \neq 0$  (generically), with winding number 0. Since the winding number is an integer-valued continuous function of  $R$ , it must jump—but it can only jump when  $P(Re^{i\theta_0}) = 0$  for some  $\theta_0$ . Therefore  $P$  has at least one root. Factor out  $(z - \alpha_1)$  and repeat.  $\square$

## 4 Proof 3: Liouville

*Proof.* If  $P$  has no roots, then  $1/P$  is entire. Since  $|P(z)| \rightarrow \infty$  as  $|z| \rightarrow \infty$  (because the leading term  $z^n$  dominates),  $1/P$  is bounded. By Liouville's theorem,  $1/P$  is constant, contradicting  $\deg(P) \geq 1$ .  $\square$

Verified numerically:  $\min_{|z|=R} |P(z)|$  grows as  $R^n$  for  $R \geq 10$ .

## 5 The circular foundation

$$\begin{array}{ccc}
 \text{Paper 91} & \longrightarrow & \text{Paper 1} \\
 \oint \frac{P'}{P} = n \quad \text{proves roots exist} & & F_P = \sum \frac{1}{z - \alpha_j} \\
 & \longrightarrow & \text{Paper 90} \\
 & & F_P \cdot P = P'
 \end{array}$$

The resolution: Paper 91 uses *only*  $P'/P$  (a ratio of known polynomials), proves that roots exist, and *then* the partial fraction decomposition  $P'/P = \sum 1/(z - \alpha_j)$  is justified.

## 6 Multiplicity

The winding integral counts roots *with multiplicity*: for  $P = (z - 1)^3(z + 1)^2$ , the integral gives  $\oint P'/P = 5 = 3 + 2$ , not 2 (the number of distinct roots).

## 7 Verification

- 500 random monic complex polynomials of degree 1–11:  $\oint P'/P dz/(2\pi i) = n$  with 0 violations.
- Topological winding of  $P(Re^{i\theta})$  around 0: equals  $n$  for  $R \geq 2$  in all test cases.
- $\min_{|z|=R} |P(z)| \rightarrow \infty$  as  $R \rightarrow \infty$ : confirmed for all test polynomials.

## References

- [1] C. F. Gauss, *Demonstratio nova theorematum*, 1799.
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- [3] I. Besevic, *The polynomial index theorem*, Pandrosion Dynamics **76** (2026).
- [4] I. Besevic, *Vieta–Newton unified: one identity*, Pandrosion Dynamics **90** (2026).